

Differential Equations

Lecture 10 — Non-Homogeneous Higher Order Linear ODEs

Non-Homogeneous Higher Order Linear ODEs

Structure of the General Solution

Definition: Non-Homogeneous Linear ODE of Order n

A differential equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = F(x), \quad (\text{I})$$

where $a_n, a_{n-1}, \dots, a_0$ and $F(x)$ are functions of x only, and $F(x) \not\equiv 0$, is called a **non-homogeneous** (or **complete**) linear ODE of order n . Setting $F(x) = 0$ gives the corresponding **homogeneous** (complementary) equation:

$$a_n \frac{d^n y}{dx^n} + \cdots + a_0 y = 0. \quad (\text{II})$$

Definition: Complementary Function and Particular Integral

Let $y_1, y_2, \dots, y_n$ be n linearly independent solutions of the homogeneous equation (II). Then:

- The **Complementary Function** (C.F.) is

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n,$$

where $c_1, \dots, c_n$ are arbitrary constants.

- A **Particular Integral** (P.I.) is any single solution y_p of (I) that contains no arbitrary constants.
- The **general solution** of (I) is

$$y = y_c + y_p.$$

Key structural fact. Every n th-order homogeneous linear ODE has exactly n linearly independent solutions. The general solution of the non-homogeneous equation always requires *both* a C.F. (to capture all degrees of freedom) *and* a P.I. (to account for the forcing term $F(x)$). Neither piece alone is sufficient.

Definition: Linear Operator

An operator L is called **linear** if, for any constants A and B and any functions $y_1(x)$ and $y_2(x)$:

$$L(Ay_1 + By_2) = A L(y_1) + B L(y_2).$$

The operator $F(D) = a_n D^n + \cdots + a_0$ is linear; this is why superposition holds.

Method of Undetermined Coefficients

The Trial Particular Integral Table

Method: Undetermined Coefficients — Choosing the Trial y_p

Step 1. Solve the homogeneous equation to find y_c .

Step 2. Read off the form of y_p from the table below, where k is the smallest non-negative integer such that *no term* in y_p already appears in y_c .

Step 3. Substitute y_p (and its derivatives) into the original ODE.

Step 4. Compare coefficients on both sides to determine the undetermined coefficients.

Step 5. Write the general solution $y = y_c + y_p$.

Form of $F(x)$	Trial form of y_p
a (constant)	$x^k A$
ax^n	$x^k (A_0 x^n + A_1 x^{n-1} + \cdots + A_{n-1} x + A_n)$
ae^{rx}	$x^k (Ae^{rx})$
$a \cos(ax)$ or $a \sin(ax)$	$x^k (A \cos(ax) + B \sin(ax))$

If $F(x)$ is a *sum* of several terms, write a separate y_p for each term and add them. The value of k must be determined independently for each term by checking against y_c .

Worked Example — Polynomial Forcing

Example: $\frac{d^2 y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x^3 - 9x^2 + 6x$

Classification. Non-homogeneous with polynomial RHS $F(x) = 2x^3 - 9x^2 + 6x$. The trial y_p is a full cubic polynomial.

Step 1 — Operator form.

$$(D^2 - 3D + 2)y = 2x^3 - 9x^2 + 6x.$$

Step 2 — Complementary function. Set the RHS to zero and form the characteristic equation:

$$\begin{aligned} m^2 - 3m + 2 &= 0 \\ (m - 2)(m - 1) &= 0 \\ m_1 &= 1, \quad m_2 = 2. \end{aligned}$$

Roots real and distinct $\Rightarrow$ Case I:

$$y_c = c_1 e^x + c_2 e^{2x}.$$

Step 3 — Choose trial y_p . Since $F(x) = ax^3$ type and no polynomial term appears in y_c , set $k = 0$:

$$y_p = Ax^3 + Bx^2 + Cx + D.$$

Step 4 — Differentiate.

$$\begin{aligned}y_p' &= 3Ax^2 + 2Bx + C, \\y_p'' &= 6Ax + 2B.\end{aligned}$$

Step 5 — Substitute into the ODE.

$$\begin{aligned}(6Ax + 2B) - 3(3Ax^2 + 2Bx + C) + 2(Ax^3 + Bx^2 + Cx + D) &= 2x^3 - 9x^2 + 6x \\2Ax^3 + (-9A + 2B)x^2 + (6A - 6B + 2C)x + (2B - 3C + 2D) &= 2x^3 - 9x^2 + 6x + 0.\end{aligned}$$

Step 6 — Compare coefficients.

$$\begin{aligned}x^3: \quad 2A &= 2 \quad \implies \quad A = 1, \\x^2: \quad -9A + 2B &= -9 \quad \implies \quad -9 + 2B = -9 \quad \implies \quad B = 0, \\x^1: \quad 6A - 6B + 2C &= 6 \quad \implies \quad 6 - 0 + 2C = 6 \quad \implies \quad C = 0, \\x^0: \quad 2B - 3C + 2D &= 0 \quad \implies \quad 0 - 0 + 2D = 0 \quad \implies \quad D = 0.\end{aligned}$$

Step 7 — Particular integral.

$$y_p = x^3.$$

Step 8 — General solution.

$$y = c_1 e^x + c_2 e^{2x} + x^3.$$

Method of Variation of Parameters

Motivation and Setup

The method of undetermined coefficients only works when $F(x)$ is a polynomial, exponential, sine, cosine, or their products — functions whose derivatives eventually repeat. For general $F(x)$ (e.g. $\tan x$, $\sec x$, $\ln x$), we need the **method of variation of parameters** (Lagrange, 1736–1813).

The method applies to the second-order non-homogeneous ODE

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = F(x), \quad (\text{III})$$

where P, Q, F are continuous on some interval I .

Derivation of the Formulas

Method: Variation of Parameters (Derivation)

Step 1. Find $y_c = c_1y_1 + c_2y_2$ by solving the homogeneous equation.

Step 2 — Vary the parameters. Replace the constants c_1, c_2 by unknown functions $u_1(x), u_2(x)$ and postulate

$$y_p = u_1y_1 + u_2y_2. \quad (\text{IV})$$

Step 3 — First-derivative condition. Differentiate (IV):

$$y'_p = (u'_1y_1 + u'_2y_2) + u_1y'_1 + u_2y'_2.$$

To prevent second derivatives of u_1, u_2 from appearing, impose:

$$u'_1y_1 + u'_2y_2 = 0. \quad (\text{V})$$

Then $y'_p = u_1y'_1 + u_2y'_2$.

Step 4 — Second derivative. Differentiate again:

$$y''_p = u'_1y'_1 + u'_2y'_2 + u_1y''_1 + u_2y''_2.$$

Step 5 — Substitute into (III). After substitution and using the fact that y_1, y_2 satisfy the homogeneous equation:

$$u'_1y'_1 + u'_2y'_2 = F(x). \quad (\text{VI})$$

Step 6 — Solve the 2×2 system (V) and (VI) by Cramer's rule:

$$u'_1 = \frac{-y_2F(x)}{W}, \quad u'_2 = \frac{y_1F(x)}{W},$$

where the **Wronskian** is

$$W = \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix} = y_1y'_2 - y'_1y_2 \neq 0.$$

Step 7 — Integrate to obtain u_1 and u_2 :

$$u_1(x) = - \int \frac{y_2 F(x)}{W} dx, \quad u_2(x) = \int \frac{y_1 F(x)}{W} dx.$$

Step 8. Substitute back into $y_p = u_1y_1 + u_2y_2$ and write $y = y_c + y_p$.

Why condition (V)? If we did not impose it, differentiating y'_p a second time would introduce u''_1 and u''_2 , giving a second-order ODE system in the unknowns — much harder to solve. Condition (V) keeps the system algebraic (first-order in u'_1, u'_2).

Worked Example — Variation of Parameters

Example: $\frac{d^2y}{dx^2} + 9y = \sec 3x$

Classification. Non-homogeneous; $F(x) = \sec 3x$ is not a polynomial, exponential, or sinusoid, so the method of undetermined coefficients does *not* apply. Use variation of parameters.

Step 1 — Operator form.

$$(D^2 + 9)y = \sec 3x.$$

Step 2 — Complementary function. Characteristic equation:

$$m^2 + 9 = 0 \implies m^2 = -9 \implies m = \pm 3i.$$

Purely imaginary roots ($p = 0, q = 3$) $\Rightarrow$ Case III:

$$y_c = c_1 \cos 3x + c_2 \sin 3x, \quad y_1 = \cos 3x, \quad y_2 = \sin 3x.$$

Step 3 — Wronskian.

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= \cos 3x \cdot (3 \cos 3x) - (-3 \sin 3x) \cdot \sin 3x \\ &= 3 \cos^2 3x + 3 \sin^2 3x \\ &= 3 \neq 0. \quad \checkmark \end{aligned}$$

Step 4 — Compute $u_1(x)$.

$$\begin{aligned} u_1(x) &= - \int \frac{y_2 F(x)}{W} dx \\ &= - \int \frac{\sin 3x \cdot \sec 3x}{3} dx \\ &= -\frac{1}{3} \int \frac{\sin 3x}{\cos 3x} dx \\ &= \frac{1}{9} \int \frac{-3 \sin 3x}{\cos 3x} dx \quad [\text{multiply numerator and denominator by } -3] \\ &= \frac{1}{9} \ln(\cos 3x). \end{aligned}$$

(Here we used the substitution $t = \cos 3x$, $dt = -3 \sin 3x dx$, so $\int \frac{-3 \sin 3x}{\cos 3x} dx = \int \frac{dt}{t} = \ln |t| = \ln |\cos 3x|$.)

Step 5 — Compute $u_2(x)$.

$$\begin{aligned} u_2(x) &= \int \frac{y_1 F(x)}{W} dx \\ &= \int \frac{\cos 3x \cdot \sec 3x}{3} dx \\ &= \frac{1}{3} \int \cos 3x \cdot \frac{1}{\cos 3x} dx \\ &= \frac{1}{3} \int dx = \frac{x}{3}. \end{aligned}$$

Step 6 — Particular integral.

$$\begin{aligned}y_p &= u_1 y_1 + u_2 y_2 \\&= \frac{1}{9} \ln(\cos 3x) \cdot \cos 3x + \frac{x}{3} \cdot \sin 3x.\end{aligned}$$

Step 7 — Apply initial/boundary conditions. None given; write the general solution directly.

Step 8 — General solution.

$$y = c_1 \cos 3x + c_2 \sin 3x + \frac{\cos 3x}{9} \ln(\cos 3x) + \frac{x \sin 3x}{3}.$$

Boundary Value Problem — Second-Order ODE

Definition: Boundary Value Problem (BVP)

A **second-order BVP** consists of a second-order ODE together with conditions specified at *two different* values of x (the boundary points), rather than at a single point (which would be an initial value problem).

Example. Solve

$$\frac{d^2y}{dx^2} + y = 0, \quad y(0) = 2, \quad y\left(\frac{\pi}{2}\right) = -3.$$

The general solution of the ODE is $y = A \cos x + B \sin x$. Applying the boundary conditions:

$$\begin{aligned} y(0) = 2 &\Rightarrow A = 2, \\ y\left(\frac{\pi}{2}\right) = -3 &\Rightarrow B = -3. \end{aligned}$$

Hence the solution of the BVP is

$$y = 2 \cos x - 3 \sin x.$$

BVP vs. IVP. In an Initial Value Problem all conditions are given at the *same* point $x = x_0$. In a BVP they are split across *two* boundary points. A BVP may have a unique solution, infinitely many solutions, or no solution — unlike IVPs with smooth coefficients, which always have a unique solution.

Summary Table

Topic	When to use	Key formula / result
C.F. + P.I. structure	Always, for any linear ODE	$y = y_c + y_p$
Undetermined Coefficients	$F(x)$ is polynomial, e^{rx} , cos/sin, or products	Trial y_p from table; set $k \geq 1$ if trial overlaps y_c
Variation of Parameters	Any continuous $F(x)$; especially $\sec x$, $\tan x$, $\ln x$	$u_1 = -\int \frac{y_2 F}{W} dx$; $u_2 = \int \frac{y_1 F}{W} dx$; $y_p = u_1 y_1 + u_2 y_2$
Wronskian W	Must be non-zero to apply V.o.P.	$W = y_1 y_2' - y_1' y_2$
BVP (2nd order)	Conditions at two boundary points	Find y_c , apply both boundary conditions to solve for constants

$F(x)$ type	Trial y_p (Undetermined Coefficients)	Overlap rule
Constant a	$x^k A$	$k = 0$ if constant not in y_c
Polynomial ax^n	$x^k(A_0 x^n + A_1 x^{n-1} + \dots + A_n)$	Increment k for each overlap
Exponential ae^{rx}	$x^k A e^{rx}$	$k =$ multiplicity of r as char. root
$a \cos(ax)$ or $a \sin(ax)$	$x^k(A \cos ax + B \sin ax)$	Both cos and sin always needed

Solution Checklist

1. Write the ODE in operator form $F(D)y = F(x)$.
2. Solve $F(D)y = 0$ to find y_c (characteristic equation $\Rightarrow$ Cases I, II, or III from Lecture 9).
3. Choose method for y_p : Undetermined Coefficients if $F(x)$ is standard; Variation of Parameters otherwise.
4. Compute y_p fully (no omitted steps).
5. Apply initial or boundary conditions if given.
6. Write $y = y_c + y_p$ and verify that the number of arbitrary constants equals the order of the ODE.